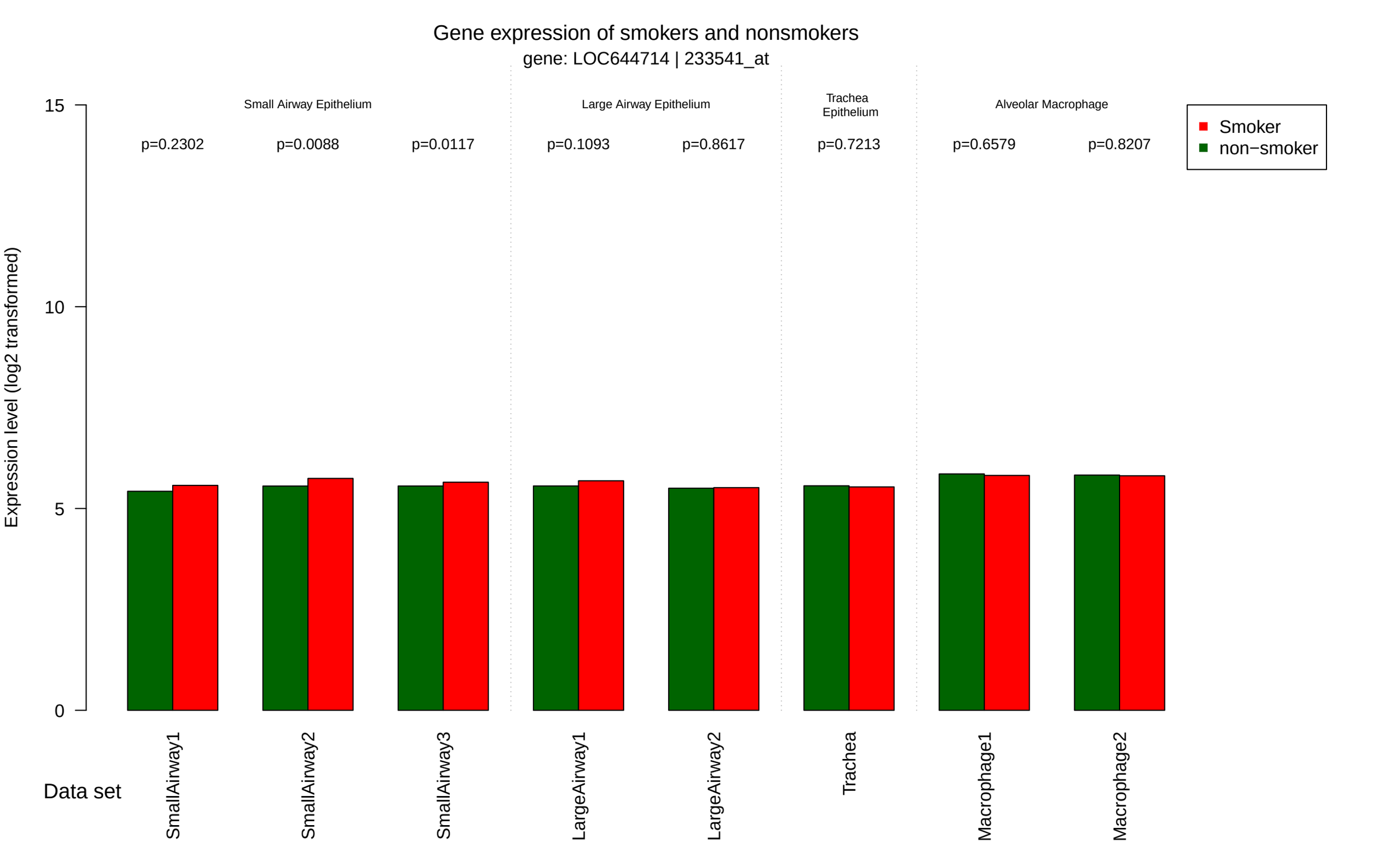
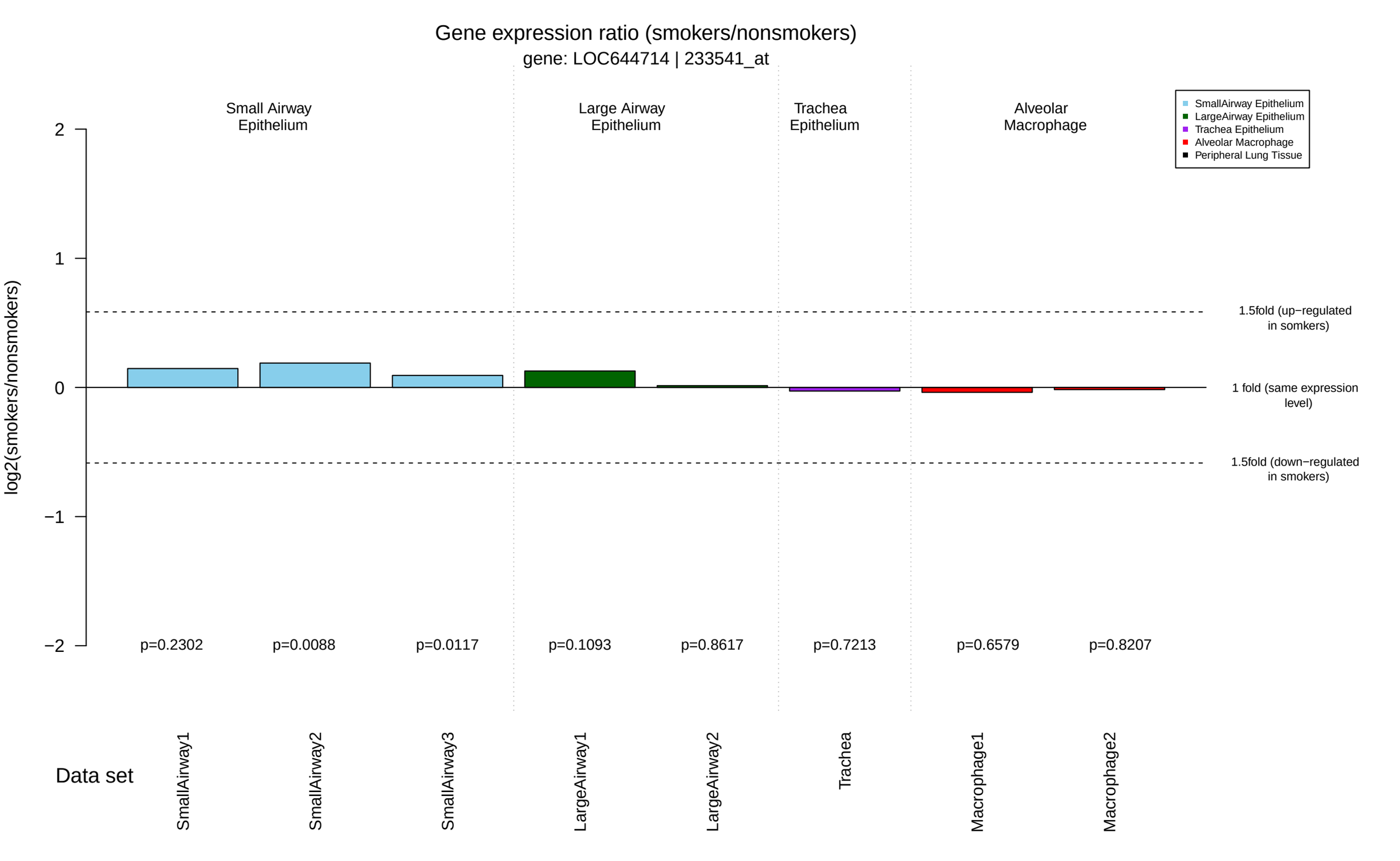
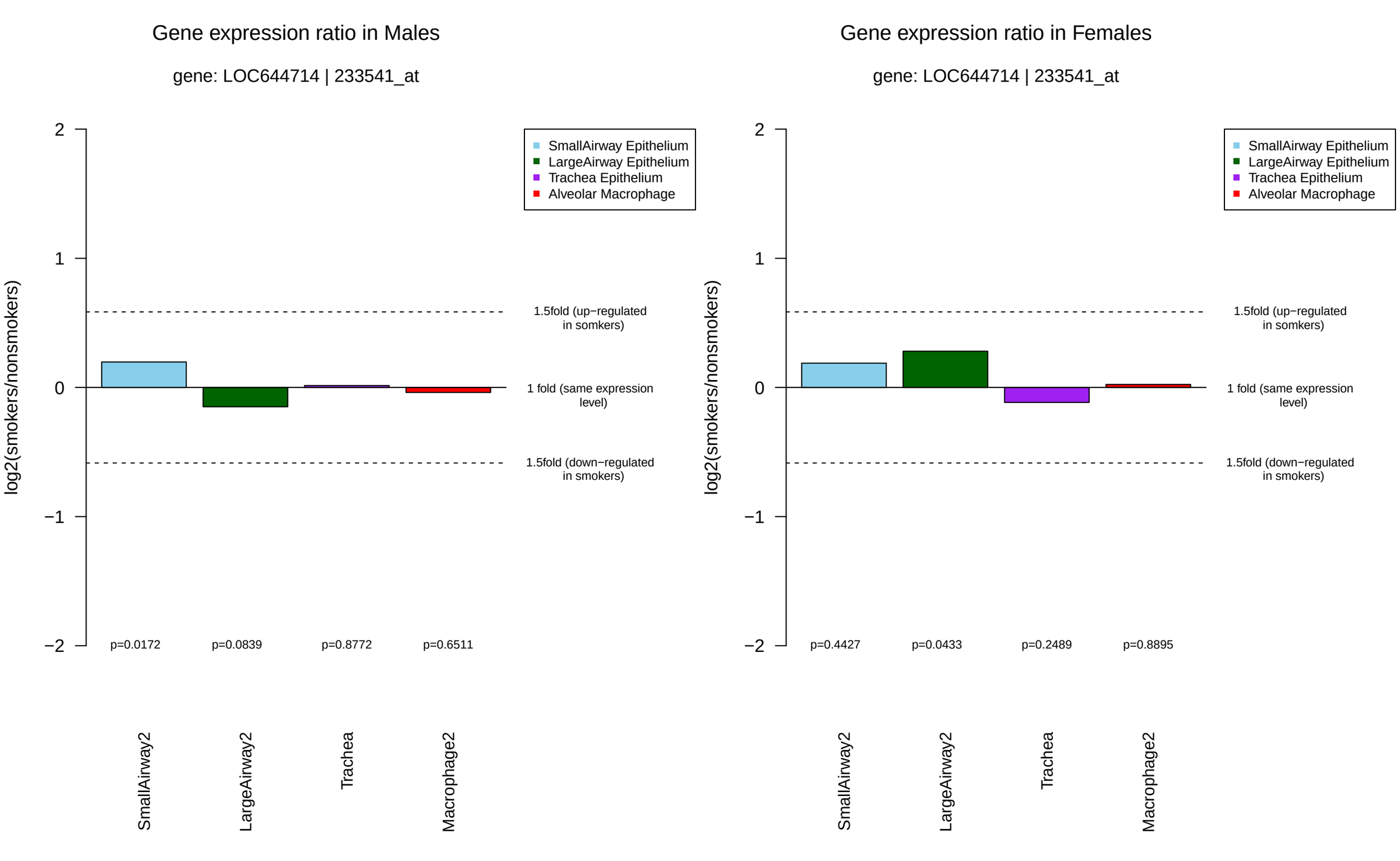
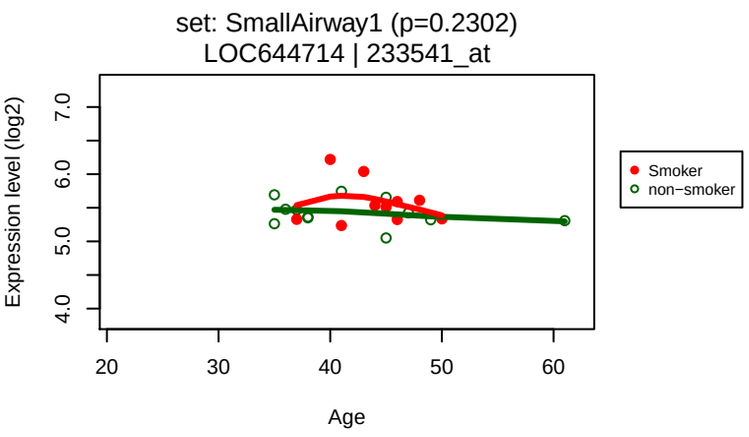
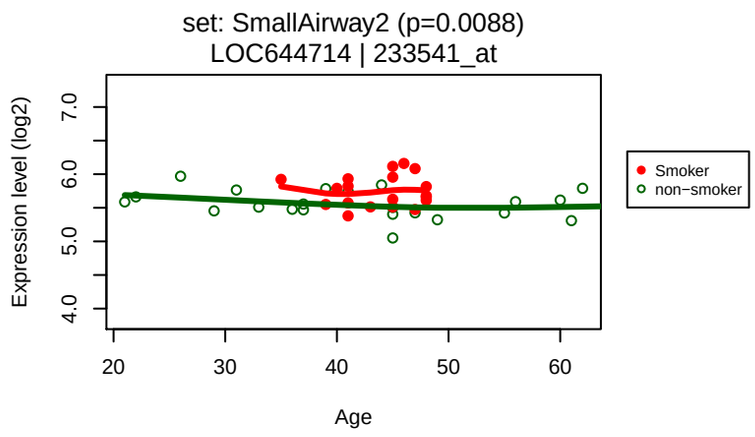
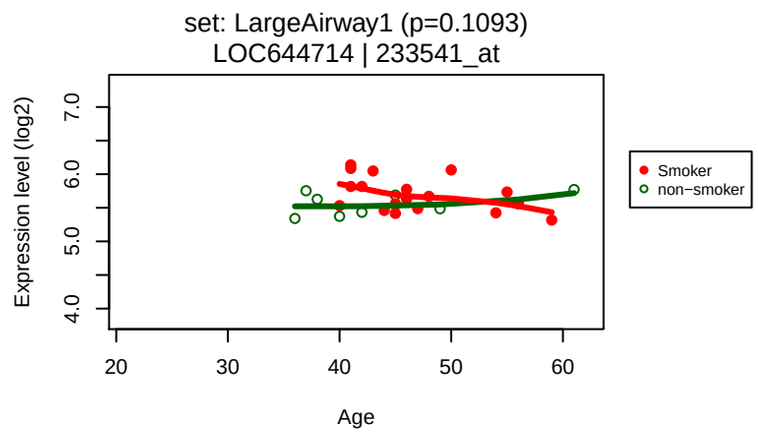

Gene expression of smokers and nonsmokers
gene: LOC644714 | 233541_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2302
p=0.0088
p=0.0117
p=0.1093
p=0.8617
p=0.7213
p=0.6579
p=0.8207
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

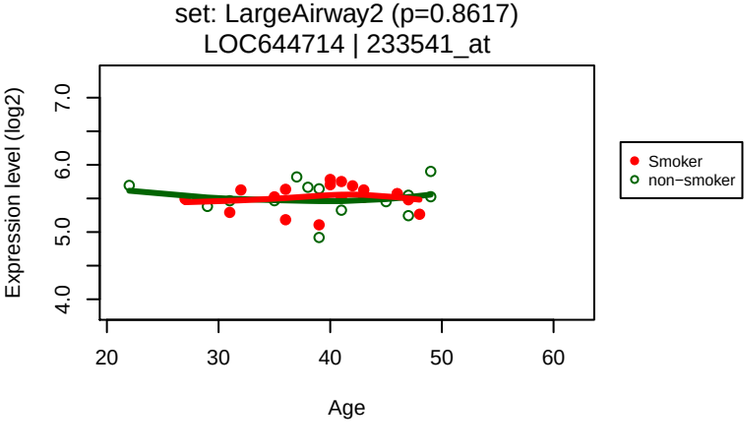

Gene expression ratio (smokers/nonsmokers)
gene: LOC644714 | 233541_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2302
p=0.0088
p=0.0117
p=0.1093
p=0.8617
p=0.7213
p=0.6579
p=0.8207
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

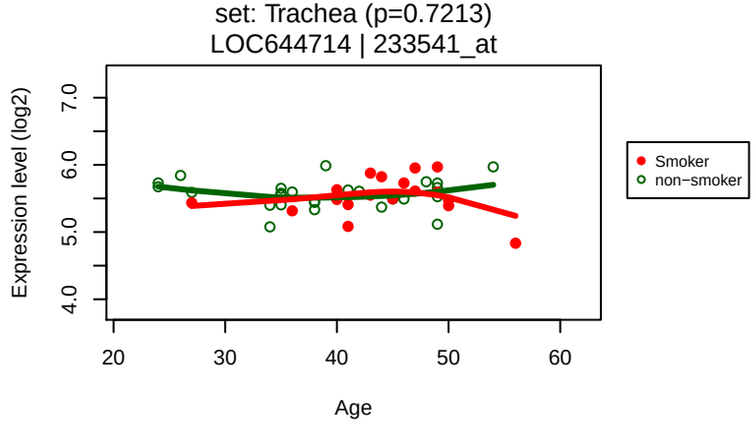

Gene expression ratio in Males
gene: LOC644714 | 233541_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0172
p=0.0839
p=0.8772
p=0.6511
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC644714 | 233541_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4427
p=0.0433
p=0.2489
p=0.8895
SmallAirway2
LargeAirway2
Trachea
Macrophage2

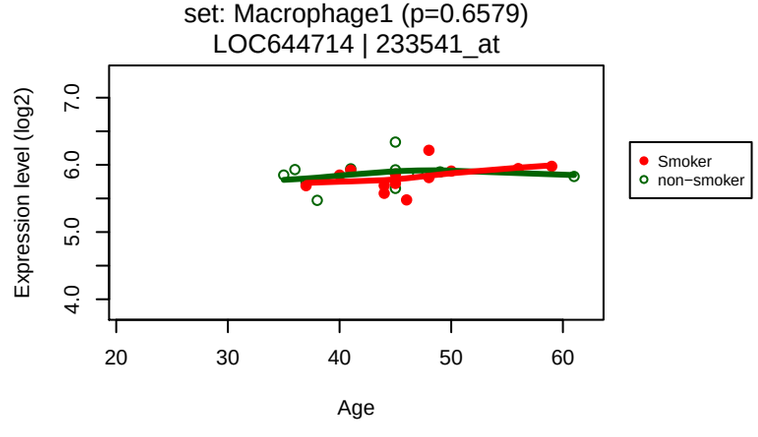

set: SmallAirway1 (p=0.2302)
LOC644714 | 233541_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker

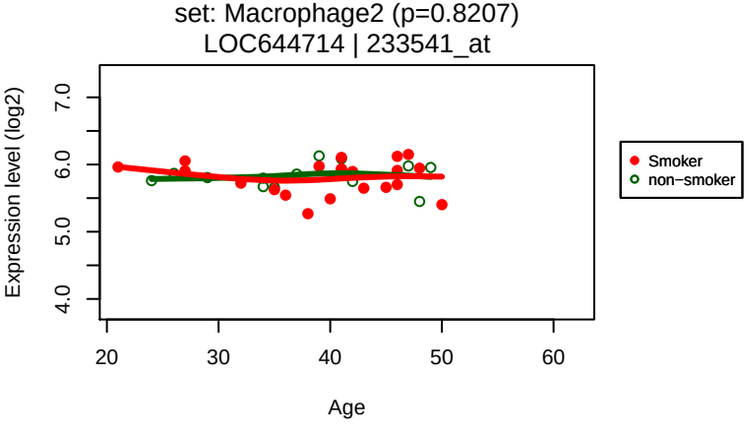

set: SmallAirway2 (p=0.0088)
LOC644714 | 233541_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker

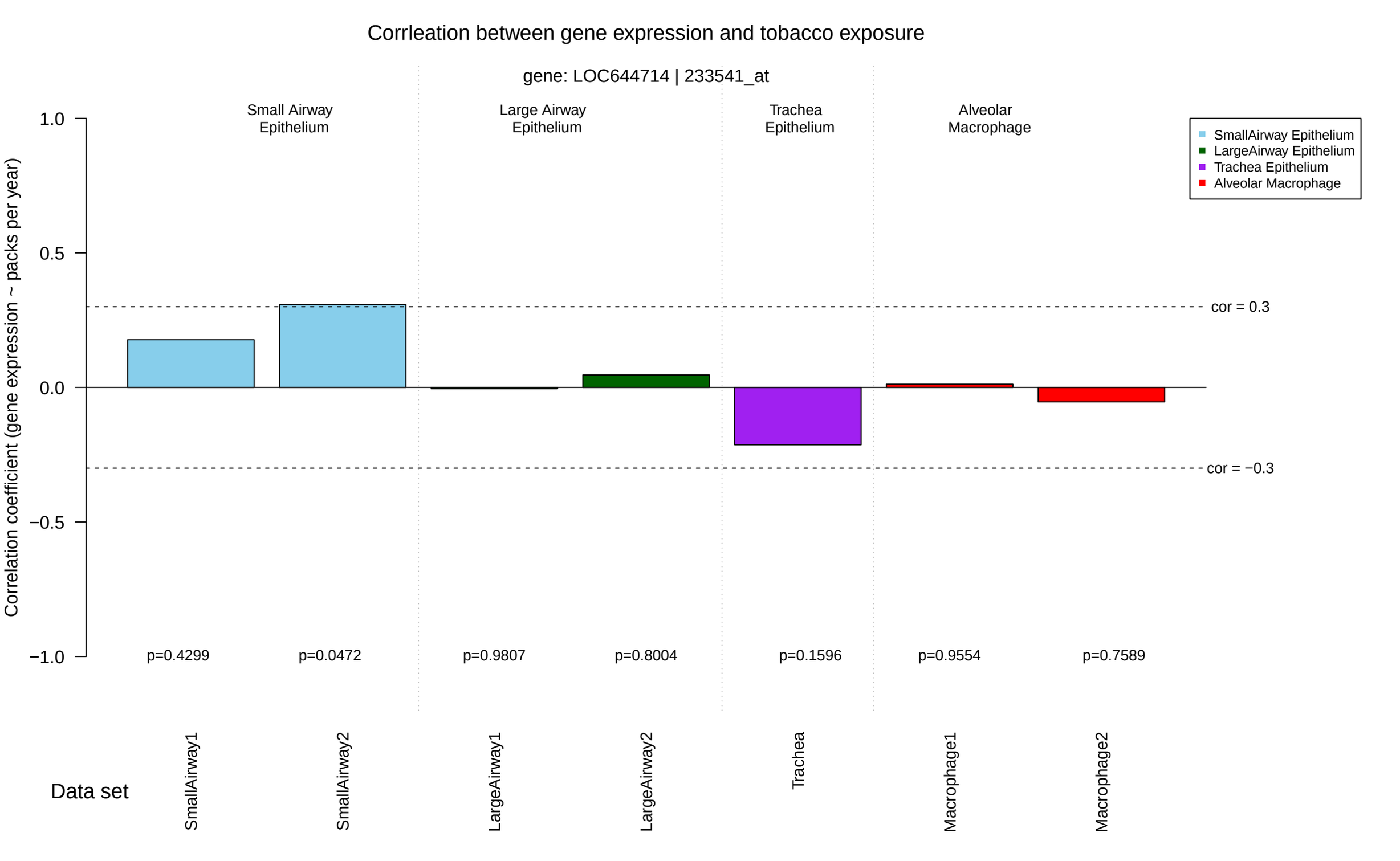

set: LargeAirway1 (p=0.1093)
LOC644714 | 233541_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8617)
LOC644714 | 233541_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.7213)
LOC644714 | 233541_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6579)
LOC644714 | 233541_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8207)
LOC644714 | 233541_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC644714 | 233541_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4299
p=0.0472
p=0.9807
p=0.8004
p=0.1596
p=0.9554
p=0.7589
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2